

WattMan: Enabling Energy-Saving And Predictive Maintenance For Businesses



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Data analytics and the Internet of Things (IoT) have successfully stepped into the realm of energy efficiency. Using these technologies, businesses with geographically distributed framework can follow a centralised predictive and preventive maintenance model for their energy-driven assets and save a great deal on expenses. Gurgaon-based startup Zenatix Solutions has developed such a solution. Called WattMan, this end-to-end IoT-based building management system (BMS) can help businesses cut their electricity consumption by up to 30 per cent and substantially mitigate their maintenance struggles.

WattMan lets businesses like retail chains, banks and cold storages identify an upcoming infrastructural malfunction and take preventive measures accordingly. For example, Mother Dairy is using WattMan to check whether the storage cooling system at its outlets is drawing excess electricity than what it should. They get the cooling system serviced while arranging for a backup before the actual breakdown happens, essentially preventing unnecessary hassles, spoilage of products and interrupted business hours.

How the idea was conceived

Zenatix was founded by IIT Delhi batchmates Amarjeet Singh, Vishal Bansal and Rahul Bhalla. The idea of working in the energy efficiency domain came from Amarjeet after his stint at the Indraprasth Institute of Information

Technology, Delhi, as a faculty member of an energy efficiency research project. He was joined by Rahul, who had worked in the legal outsourcing industry, and Vishal, who was associated with ING Global since his graduation from IIM Ahmedabad.

However, the road was not smooth as they had to take detours in their development to get the expected results in the form of WattMan. "We started off trying to become an energy analytics company. However, to provide the analytics we needed to collect the data. For that, the devices needed to be connected. So we had to become an IoT company first. We had to build a robust hardware that would allow us to collect a large volume of data at any application scale," said Amarjeet. It took about one-and-a-half year for the Zenatix team to complete WattMan.

The intelligence behind WattMan

WattMan BMS, driven by technologies like IoT and artificial intelligence (AI), provides intelligent controls together with predictive and preventive maintenance. With WattMan, enterprises can ensure that all electrical appliances are switched on/off automatically according to an intelligent schedule, eventually resulting in no wastage of electricity. Additionally, businesses can prepare before any appliance breaks down.

The complete WattMan architecture consists of four main layers: sensing layer, edge layer, an IoT stack over the cloud and an AI suite.

The lowest layer of the stack consists of sensors and control points. To serve different business requirements, only this sensing layer has to be customised, keeping the rest of the architecture intact. Edge devices were designed with great effort and care to comply with the local environmental conditions.

Amarjeet says, "Connecting the infrastructure locally in the store as well as to the Internet is the difficult part. Given the local challenges in India, the control



WattMan kit with energy meter



WattMan Lite

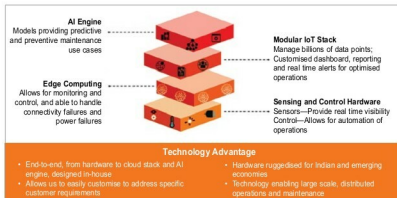


Fig. 1: The complete WattMan architecture